# Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · Agentic Workflows · LangGraph · LLM Evaluation Frameworks · RAG Pipelines · AWS Bedrock · Anthropic · Claude Code · PropTech · Python · Production AI

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

## PROFESSIONAL SUMMARY

Senior AI Engineer with two production agentic systems that map directly to Crexi's AI platform needs. AgentProbe is an open-source LLM evaluation framework - 5-metric quality engine, LLM-as-judge scoring, CI-integrated regression gate, and multi-provider Anthropic/OpenAI/Ollama routing - built on the principle that evaluation infrastructure comes before shipping prompts. LeaseI is an 8-agent LangGraph stateful system with multi-step agentic workflows (routing, planning, tool use, state management), multi-source RAG retrieval, deterministic guardrails on regulated outputs, HITL escalation on low-confidence decisions, and source-linked answers with 100% citation correctness - 93.6% precision across 1,240 compliance test cases. Leverages Claude Code and agentic coding tools daily in professional workflows. AWS infrastructure background (S3, EC2, RDS, Glue) with Anthropic API production experience. 8 years production Python engineering at Entrata: 99.5% uptime, 100K+ MAU, RabbitMQ event-driven systems, and observability tooling. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**AI Engineering:** Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, LLM Evaluation Frameworks, LLM-as-Judge, CI-Integrated Regression Gating, Offline Evals, HITL, Guardrails, Structured Outputs, Source-Linked Generation, Prompt Engineering, ChromaDB, Vector Search, Embeddings, Anthropic API (Claude), OpenAI API, AWS Bedrock-compatible, Groq, Ollama, Claude Code, MCP, BERT, spaCy, NLP

**Engineering & Production:** Python (8yr), TypeScript, JavaScript, FastAPI, SQL, Docker, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Bash, Linux, Git

**Observability & Data:** Production Monitoring, Reliability Engineering, Observability, Performance Engineering, PHI Compliance, PyTorch, scikit-learn, NumPy, Pandas, PySpark, Feature Engineering, Statistical Modeling, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Designed and implemented multi-step agentic workflows (LangGraph) for mission planning - routing, planning, and tool use with RAG retrieval over distributed data; deterministic guardrails and structured outputs before delivery; CI/CD-integrated evals gating every release.
- Leveraged Claude Code and agentic coding tools throughout the development lifecycle; implemented HITL escalation for edge cases requiring human review; observability on retrieval quality and latency.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python and data science; designed monitoring dashboards improving research comprehension by 40%.

### Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Production AI and data pipeline ownership at scale** - owned HUD income-limit pipeline end-to-end for 7 years (federal source → 180,000 domain records powering eligibility and compliance for 100K+ housing units); AI-driven observability and monitoring tooling reducing triage time 40%; 99.5% uptime SLA at 100K+ monthly active users.
- **Agentic coding and production Python infrastructure** - leveraged AI-assisted development workflows to scaffold, debug, and maintain distributed Python systems; RabbitMQ event-driven async processing with retry and idempotency guarantees; Redis caching; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS at enterprise scale.
- **Engineering standards and observability** - 200+ SQL query optimizations (30% gain); 100% PHI compliance with Libsodium-encrypted pipelines; 300+ code reviews; CI/CD via GitHub Actions; mentored 5+ engineers on production AI and backend best practices.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### AgentProbe: Open-Source LLM Evaluation Framework

Evals · Production Monitoring

Python · FastAPI · LangGraph · ChromaDB · CI/CD · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation framework with offline evals and CI-integrated regression gating: 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, multi-provider Anthropic/OpenAI/Ollama routing for cost optimization. Evaluation infrastructure built before prompts - zero regressions shipped across 1,240+ test cases. Open-source contribution to AI tooling.

### Leasel: 8-Agent Agentic Workflow Platform with Source-Linked RAG

Agentic · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system: 8-agent LangGraph stateful workflow with multi-step routing, planning, tool use, and state management; multi-source RAG retrieval pipeline for proprietary compliance data; deterministic guardrails on every regulated output; HITL escalation when model confidence below threshold; source-linked answers with 100% citation correctness. 93.6% precision, 87.9% F1 across 1,240 compliance test cases; CI quality gate with full observability.

### AI on FHIR: NLP-Driven Medical Records Query System

RAG · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

RAG pipeline with context and retrieval over proprietary structured records: BERT embeddings + LLM inference translating natural-language queries into FHIR API calls - reduced retrieval time ~60%; source-linked outputs with HIPAA-compliant RBAC and full data isolation.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017